Commodities Handbook

Specialist reference across energy, metals, agriculture, softs, livestock and environmental markets

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Foundations

Market Structure & Pricing Mechanics

This chapter sets out the toolkit used in every later chapter: how futures curves are formed, what spreads mean, and how physical and paper markets connect. Commodity-specific chapters assume this vocabulary.

Cost of carry and the forward curve

For a storable commodity, the theoretical forward price is bounded by the **full carry**:

> **F = S × e^((r + u − y) × T)** where *r* = financing rate, *u* = storage + insurance cost, *y* = convenience yield, *T* = time. Contango can never sustainably exceed full carry, because a trader could buy spot, store it, sell forward and lock in a riskless profit. Backwardation has **no theoretical limit**, because you cannot borrow physical barrels from the future.

- **Contango** (deferred > prompt): ample inventories, low convenience yield. Deep contango approaching full carry means storage is filling and the market is paying someone to hold stock.
- **Backwardation** (prompt > deferred): scarce prompt supply, high convenience yield. Inventories and spreads are inversely related; the inventory–spread relationship (the "Working curve") is one of the most reliable relationships in commodities.
- **Convenience yield** is the implied benefit of holding the physical: avoiding a plant shutdown, meeting a contract, keeping a refinery running.

Spreads: the language of fundamentals

Spread type	Example	What it tells you
Calendar / time	CL Dec–Jan, ZC Jul/Dec	Prompt tightness, storage economics, old-crop vs new-crop
Location	WTI–Brent, LME vs SHFE copper, KC–Chicago wheat	Freight, arbitrage windows, regional balance
Quality	Brent–Dubai, Arabica–Robusta, 62%–58% Fe ore	Sweet/sour, grade premiums, demand by product
Processing / margin	3-2-1 crack, soy crush, spark/dark spread, hog crush	Margin of the converter; predicts run rates
Inter-commodity	Gold/silver ratio, corn/wheat, soyoil/heating oil (BOHO)	Substitution and relative value

Rule of thumb: outright prices tell you *what* the market thinks; spreads tell you *why*. A rally on falling spreads is usually speculative or macro-driven; a rally led by the front spread is usually physical tightness.

Basis and physical pricing

- **Basis = local cash price − futures price.** Physical deals are typically priced as futures + basis (grains: "Dec + 30 over"), or index + differential (crude: "Dated Brent + $1.20").
- Hedgers trade **basis risk** for flat-price risk. A grain elevator long cash/short futures profits when basis strengthens.
- Basis drivers: local supply/demand, freight, storage, quality, delivery logistics, and the convergence mechanism of the contract.
- **EFP / EFRP** (exchange for physical / related position) links an OTC physical deal to a futures position and is how much physical volume enters the exchange.

Delivery and settlement

- **Physically delivered** contracts (WTI, Henry Hub, CBOT grains, COMEX metals, LME metals, ICE softs) force convergence through delivery; watch first notice day (FND) and the delivery mechanics, as positions must be rolled or offset beforehand.
- **Cash-settled** contracts (Brent, feeder cattle, lean hogs, iron ore SGX, JKM) settle against an index; convergence depends on index integrity.
- **Squeezes** happen when deliverable supply is small relative to open interest near expiry. Classic cases: LME nickel March 2022, Hunt brothers silver 1980, WTI April 2020 negative price (storage full at Cushing).

Roll yield and index economics

- Total return of a futures position = spot return + **roll yield** + collateral yield.
- Long-only index funds (GSCI, BCOM) roll in a published window (BCOM / GSCI around the 5th–9th business day). The "Goldman roll" can pressure the front spread predictably.

- Backwardated markets pay positive roll yield to longs; contango markets bleed.

Positioning and flows

- **CFTC Commitments of Traders** (Fridays, data as of Tuesday): Disaggregated report splits Producer/Merchant, Swap Dealers, Managed Money and Other Reportables. Extreme managed-money positioning is a contrarian signal, but positioning can stay extreme for long periods.
- **ICE Europe COT** covers Brent, gasoil, London softs; **LME COTR** covers base metals.
- CTAs (trend followers) amplify moves; know the moving-average levels where their signals flip.
- Options: open interest at round strikes creates pinning and gamma-driven hedging flows near expiry.

Volatility characteristics

- Commodity volatility skew is often **call-skewed** (upside fear) in supply-shock markets — ags in summer weather season, natural gas into winter, cocoa/coffee in deficits — unlike equities' put skew.
- Seasonal implied volatility: corn/soy implied vol rises into the June–August pollination window and collapses after the August WASDE once yield is clearer.
- Mean-reversion in spreads, trend in outrights is a useful heuristic but not a law.

Hedging basics

1. **Producer hedge:** short futures/forwards or buy puts; collar (buy put, sell call) to cut premium.
2. **Consumer hedge:** long futures or buy calls; airlines hedge jet via Brent/heating oil (cross-hedge with basis risk).
3. **Hedge ratio** = Cov(ΔS, ΔF) / Var(ΔF); cross-hedges need an empirically estimated ratio.
4. Margin calls on hedges can create liquidity crises even when the hedge is economically correct (Metallgesellschaft 1993).

Producer and consumer flows and the curve

Hedging flows mostly move the **back** of the curve, while inventories set the **front**: *"the front is physical, the back is financial."*

- **Producers sell forward** (oil and gas companies, farmers, miners), usually 6–36 months out. This weighs on deferred prices: backwardation steepens and contango flattens.
- **Consumers buy forward** (airlines, utilities, food processors, fabricators), lifting deferred prices. Consumers usually hedge shorter-dated and more opportunistically than producers.
- **Speculators take the other side** and are paid for it. Keynes's *normal backwardation*: when producer hedging dominates, deferred futures trade below the spot price expected later, and the gap is the speculator's risk premium. It reverses when consumers dominate.

Flow	When it happens	Effect on the curve
Producer hedging	Into rallies: producers sell at budget or breakeven levels	Caps the back; steepens backwardation in a rally ("hedging wall")
Producer hedge buybacks	Into selloffs: hedges unwound or restructured	Supports the back; flattens the curve
Consumer hedging	On dips: buyers lock in cheap costs	Lifts the back; flattens backwardation
Consumer panic buying	After supply shocks, when buyers are underhedged	Front and back rally together; whole curve shifts up
Storage plays	When contango covers storage and financing	Buy front, sell back; caps contango at full carry
Long-only index rolls	Monthly roll window	Weakens the front spread during the window

Producer hedging is **pro-cyclical**: it sells strength. That is why long-dated prices are far less volatile than prompt prices, and why a supply-shock rally lifts the front much more than the back.

Options flows

- Producers often hedge with **collars** (buy puts, sell calls). Dealers on the other side end up **short downside puts**, which they hedge by selling futures as prices fall, so selloffs accelerate. They are also **long upside calls**, which they hedge by selling into rallies, so rallies are dampened.
- The result is put-heavy skew in producer-dominated markets and fast, air-pocket selloffs. **Mexico's annual oil hedge** is the classic example: a very large put purchase that dealers hedge in the deferred months.

By market

- **Crude oil:** US shale producers hedge heavily 12–24 months out, historically anchoring long-dated WTI near the cost of new shale supply. Airlines are the main consumer hedgers, but irregular. When producers are underhedged, rallies can lift the back end hard.
- **Natural gas:** producers sell the 12-month strip; utilities and local distribution companies buy winter months ahead, supporting the summer–winter spread.
- **Grains:** farmer selling drives harvest pressure on new-crop contracts and weaker basis. Farmers holding grain keep the old-crop/new-crop spread firm. Elevators carry most of the commercial short.
- **Metals:** most miners no longer hedge. The unwinding of gold producers' hedge books in the 2000s removed a major seller from the back of the curve. Fabricators and cable makers are the main back-end buyers.
- **Softs:** origin selling caps rallies (Brazilian coffee farmers on a weak BRL, West African cocoa boards selling forward). A board defaulting on forward sales can invert the curve fast.

Reading it in the data

1. **CFTC COT:** Producer/Merchant shorts and longs show hedging intensity; Swap Dealer positions show OTC hedges passed through to futures.
2. **Deferred-month open interest:** rising OI on rallies means producers are hedging.
3. **Option skew:** deepening put skew in deferred months means producers are buying protection.
4. **Company hedge books** (earnings disclosures): low hedge ratios leave the back end vulnerable to a squeeze.
5. **Curve vs inventories:** when the curve shape doesn't match what inventories imply, flows are usually the reason.

Rules of thumb: backwardation plus low inventories is a genuine physical shortage. A back end that falls during a front-led rally is hedging, not bearish fundamentals. When producers stop hedging (low prices, depleted hedge books), the back end tends to be well supported.

Desk notes

Add your own observations here. Anything you write in this file appears in the PDF on the next build.

Month Codes, Calendars & Key Reports

Futures month codes

Code	Month	Code	Month	Code	Month
F	January	K	May	U	September
G	February	M	June	V	October
H	March	N	July	X	November
J	April	Q	August	Z	December

Example: **CLZ6** = WTI crude December 2026; **ZSX6** = soybeans November 2026.

Recurring data that moves markets

Release	Publisher	Frequency / timing	Markets
Weekly Petroleum Status Report	US EIA	Wednesday 10:30 ET	Crude, products
API inventory	American Petroleum Institute	Tuesday 16:30 ET	Crude, products
Natural Gas Storage Report	US EIA	Thursday 10:30 ET	Henry Hub
Oil Market Report	IEA	Monthly, mid-month	Oil
Monthly Oil Market Report	OPEC	Monthly, mid-month	Oil
Short-Term Energy Outlook	US EIA	Monthly	Energy
Baker Hughes rig count	Baker Hughes	Friday 13:00 ET	Oil, gas
WASDE	USDA	Monthly, ~10th–12th, 12:00 ET	Grains, oilseeds, cotton, sugar, livestock
Crop Progress	USDA NASS	Mondays 16:00 ET (Apr–Nov)	Grains, cotton
Export Sales	USDA FAS	Thursday 08:30 ET	Grains, oilseeds, cotton, meats
Prospective Plantings / Acreage	USDA NASS	End-March / end-June	Grains, oilseeds
Quarterly Grain Stocks	USDA NASS	Quarterly (Mar, Jun, Sep, Jan)	Grains
Hogs and Pigs	USDA NASS	Quarterly	Lean hogs
Cattle on Feed	USDA NASS	Monthly	Cattle
CONAB crop reports	Brazil CONAB	Monthly	Soy, corn, coffee, sugar
UNICA cane crush	UNICA	Bi-weekly (Apr–Nov)	Sugar, ethanol
MPOB palm oil data	Malaysian Palm Oil Board	Monthly, ~10th	Palm, veg oils
Ivory Coast cocoa arrivals	Exporter estimates	Weekly	Cocoa
LME / COMEX / SHFE stocks	Exchanges	Daily (SHFE weekly)	Metals
China trade data	GACC	Monthly	Everything, esp. ore, copper, soy, crude
China PMIs	NBS / S&P Global	Monthly	Metals
CFTC Commitments of Traders	CFTC	Friday 15:30 ET	All US futures
FOMC, US CPI, payrolls	Fed / BLS	Scheduled	Gold, dollar-sensitive complex

Seasonal calendar overview

Period	Typical focus
Jan–Feb	South American soy harvest begins; US winter gas draws peak; Chinese New Year metals destock
Mar–Apr	USDA Prospective Plantings; Brazil safrinha corn planting; refinery maintenance ends; LME Week Asia
May–Jun	US planting/emergence; Atlantic hurricane season opens (1 Jun); driving season begins
Jul–Aug	Corn pollination (July) and soybean pod-set (August) — peak weather premium; peak US power burn
Sep–Oct	US harvest; Brazil soy planting; West African cocoa main crop starts (Oct); LME Week London
Nov–Dec	Gas withdrawal season starts; Southern Hemisphere wheat harvest; index rebalancing in January

Desk notes

Add calendar items, local holidays, or report reactions here.

Energy

Crude Oil

The world's largest commodity market by value. Around 100+ million barrels per day (mb/d) are consumed globally. Price formation runs through three benchmark families: **Brent** (Atlantic Basin / waterborne), **WTI** (US inland and US Gulf exports), and **Dubai/Oman/Murban** (Middle East crude into Asia).

Contract specifications

Contract	Venue / ticker	Size	Tick	Settlement	Notes
WTI Light Sweet	NYMEX CL	1,000 bbl	$0.01 = $10	Physical, Cushing, OK	Expires 3 business days before the 25th calendar day of the month before delivery
Brent	ICE B	1,000 bbl	$0.01 = $10	Cash vs ICE Brent Index (EFP available)	Expires last business day of the second month before delivery
WTI Houston (MEH)	NYMEX / ICE	1,000 bbl	$0.01	Physical / differential	US Gulf export pricing
Oman	GME (ex-DME)	1,000 bbl	$0.01	Physical	Sets Oman OSP
Murban	ICE Futures Abu Dhabi	1,000 bbl	$0.01	Physical, Fujairah	ADNOC prices Murban off this
Shanghai crude	INE SC	1,000 bbl	¥0.1	Physical, medium-sour basket	Yuan-denominated

Dated Brent (Platts) is the physical benchmark for roughly two-thirds of world crude. Basket: Brent, Forties, Oseberg, Ekofisk, Troll, and since mid-2023 **WTI Midland** delivered into Rotterdam, which now frequently sets the Dated price. The complex runs Dated → Forward (cash BFOETM) → ICE futures, linked by CFDs (contracts for difference) and EFPs.

The Brent basket: deliverable grades

"Brent" today is not one crude but a basket of light sweet grades. Platts assesses each grade for cargoes loading 10 days to one month ahead, adjusts for quality, and **the most competitive (cheapest) grade sets Dated Brent**. The same grades can be delivered against forward (cash) Brent contracts, which is what ICE Brent futures ultimately settle against.

Grade	Origin / loading	Approx. API / sulfur	Role in the basket
Brent	UK, Sullom Voe (Shetland)	~38° / ~0.4%	The original grade; now tiny volumes
Forties	UK, Hound Point (Forth) via the Forties Pipeline System	~40° / ~0.6–0.8%	Highest sulfur in the basket (Buzzard field); carries a sulfur de-escalator; historically set Dated most often
Oseberg	Norway, Sture	~37° / ~0.2–0.3%	Higher quality; earns a quality premium, so rarely sets Dated
Ekofisk	Norwegian fields, landed at Teesside (UK)	~38° / ~0.2%	Quality premium; rarely sets Dated
Troll	Norway, Mongstad	~35–36° / ~0.1–0.2%	Added in 2018; quality premium
WTI Midland	US Permian, assessed delivered (CIF) Rotterdam	~40–44° / ~0.1–0.2%	Added for June 2023 deliveries; by far the largest volume, and now sets Dated most of the time

(API and sulfur figures are approximate and drift as field mixes change; check the latest Platts methodology.)

How quality adjustments work

- **Quality premiums (escalators):** sellers of the better-quality grades (Oseberg, Ekofisk, Troll) receive a premium when delivering against a forward Brent contract, so they compete on a quality-adjusted basis.
- **Sulfur de-escalator:** Forties sellers pay the buyer a discount for each increment of sulfur above a threshold. Watch the Forties sulfur level when Buzzard output changes.
- **Freight normalisation for WTI Midland:** Midland is priced delivered into Rotterdam and netted back to a North Sea basis using transatlantic Aframax freight. A move in that freight rate changes Midland's competitiveness, and with it Dated Brent itself.

Why the grade mix matters to traders

- **Liquidity and squeeze risk:** North Sea output has been declining for decades. Each grade was added (Forties/Oseberg 2002, Ekofisk 2007, Troll 2018, Midland 2023) to keep enough deliverable cargoes in the basket to resist squeezes. Thin months can still be squeezed by one buyer accumulating cargoes.
- **US export economics now drive Dated:** with Midland usually the cheapest grade, Dated Brent is effectively linked to Permian output, the WTI–Brent arbitrage, Gulf Coast export capacity and transatlantic freight.
- **Spreads to watch:**
  - Grade differentials: Forties vs Dated, Midland vs Dated, Ekofisk/Oseberg premiums.
  - **CFDs** (Dated vs forward Brent, weekly) and **DFL** (Dated-to-frontline): the market's measure of how tight physical cargoes are compared with paper.
  - **North Sea "window" activity:** bids and offers in the Platts Market-on-Close window. A single trader buying many cargoes is a classic early sign of a squeeze.
- **Possible future additions:** Platts has consulted on adding further grades (for example, Norway's Johan Sverdrup, a medium sour crude). A sour grade in the basket would change Dated's quality and its relationship with Dubai.

Quality fundamentals

- **API gravity** (lightness) and **sulfur** (sweet <0.5%, sour >0.5%) drive refining value.
- Light sweet yields more gasoline/diesel; heavy sour needs complex refineries (cokers, hydrocrackers) but is cheaper.
- Key differentials: **Brent–Dubai EFS** (sweet/sour, Atlantic/East arbitrage); **WTI–Brent** (US export arb, normally set by freight + pipeline costs from Cushing/Midland to the Gulf); **WCS–WTI** (Canadian heavy discount, eased since the Trans Mountain Expansion started in May 2024).

Supply

Group	Approx. share	Notes
United States	~20% of liquids	Permian-led shale; short-cycle, price-responsive with ~6–9 month lag; consolidation has increased capital discipline
OPEC+ (OPEC + Russia, Kazakhstan, Oman, etc.)	~50%	Managed supply via quotas; Saudi Arabia is the swing producer with most spare capacity
Saudi Arabia	~10%	Sets monthly OSPs for Asia/Europe/US; watch Arab Light OSP to Asia
Russia	~10%	Sanctions/price cap since Dec 2022; redirected to India/China; shadow fleet
Brazil, Guyana, Canada, Norway	growing non-OPEC	Guyana (Stabroek) and Brazil pre-salt are key growth sources

- **Spare capacity** (mostly Saudi, UAE, Kuwait) is the market's shock absorber; low spare capacity = high geopolitical premium.
- **Decline rates** of existing fields average 4–8% a year; shale wells decline much faster (60–70% in the first year).
- **Breakevens:** marginal shale wells and fiscal breakevens of Gulf states both act as soft floors/ceilings for policy decisions.

Demand

- Transport fuels around 60% (gasoline, diesel, jet), petchem feedstocks rising (naphtha, LPG, ethane).
- Growth is led by non-OECD Asia (India, China), the Middle East and aviation; China's growth is slowing as EVs and LNG trucks displace gasoline and diesel.
- **Refinery runs** are the real-time demand signal; watch seasonal maintenance (spring and autumn turnarounds).

Inventories and term structure

- OECD commercial stocks vs 5-year average is the classic balance gauge; it is inversely correlated with Brent time spreads.
- **Cushing stocks** drive WTI front spreads; below ~20 mb operational minimums triggers sharp backwardation.
- Floating storage economics: contango must cover freight (VLCC time charter) + financing.
- Watch **Brent M1–M2**, **M1–M6**, **Dec–Dec** (Dec-red-Dec) spreads as balance barometers.

Key price drivers

1. OPEC+ policy (monthly/biannual meetings; voluntary cuts and unwinds).
2. Geopolitics: Strait of Hormuz (~20% of global oil flows), Red Sea/Bab el-Mandeb, Russia/Ukraine, Libya outages, Iran/Venezuela sanctions.
3. US shale productivity and rig/frac-spread counts.
4. Chinese import quotas, stockpiling (SPR), and teapot refinery margins.
5. USD and macro risk sentiment; recession probabilities.
6. Refining margins pulling or pushing crude demand.
7. SPR releases/refills (US SPR policy).

Seasonality

- Q2–Q3: northern-hemisphere driving season + Middle East crude burn for power → stronger demand.
- Spring and autumn refinery maintenance soften crude demand.
- Hurricane season (Jun–Nov) affects US Gulf production and refining.

Trading notes

- Use **crack spreads** to judge whether crude strength is demand-pulled (cracks strong) or supply-pushed (cracks weak).
- Watch **Dubai spreads and Brent–Dubai EFS** for Asian demand and OPEC+ sour supply.
- Timespreads usually lead flat price at turning points.
- Big moves often occur around OPEC+ meetings, EIA Wednesdays, and geopolitical headlines; options skew flips to calls during supply scares.

Desk notes

> Add your own observations here.

Refined Products

Refinery margins connect crude to consumers. Products often lead crude at turning points because they reflect end-user demand.

Contract specifications

Contract	Venue / ticker	Size	Tick	Delivery
RBOB Gasoline	NYMEX RB	42,000 gal	$0.0001/gal = $4.20	NY Harbor
NY Harbor ULSD (heating oil)	NYMEX HO	42,000 gal	$0.0001/gal = $4.20	NY Harbor
Low Sulphur Gasoil	ICE G	100 t	$0.25/t = $25	ARA barges (Amsterdam–Rotterdam–Antwerp)
Singapore gasoil, fuel oil, naphtha	Swaps (Platts)	various	—	Cash-settled vs Platts assessments

Conversion: **1 tonne gasoil ≈ 7.45 bbl**; 1 bbl = 42 US gallons.

Refining basics

- **Simple (hydroskimming)** refineries distil crude; **complex** refineries add FCC, hydrocrackers and cokers to convert heavy residue into light products.
- Typical US Gulf yields: ~45% gasoline, ~30% distillate, ~10% jet, remainder LPG, resid, coke.
- **Nelson Complexity Index** measures refinery sophistication.

Margins

Spread	Formula	Use
3-2-1 crack	(2 × RBOB + 1 × HO) − 3 × WTI, per bbl	US Gulf/Midcon refinery margin proxy
Gasoline crack	RBOB ($/gal × 42) − crude	Driving season demand
Diesel crack	HO ($/gal × 42) − crude, or ICE gasoil − Brent	Industrial/freight demand
Gasoil–Brent	Gasoil ($/t ÷ 7.45) − Brent	European middle distillates
Singapore complex margin	Weighted product basket − Dubai	Asian refining

Gasoline

- US is ~45% of global gasoline demand; peak demand Memorial Day–Labor Day.
- **RVP seasonality:** summer-grade (low vapour pressure) gasoline from April onwards costs more; RBOB contract months switch spec (April–September contracts are summer grade). This creates a jump in the Mar/Apr spread.
- Ethanol blending (~10% in US) links RBOB to corn and RINs (D6).

Distillates (diesel, gasoil, heating oil, jet)

- Most economically sensitive barrel: trucking, rail, agriculture, mining, shipping, heating.
- Europe is structurally short diesel and historically imported from Russia; post-2023 EU embargo, it imports from the US Gulf, Middle East and India.
- Winter heating demand (US Northeast, Europe) and low stocks cause sharp backwardations.
- **IMO 2020** (0.5% sulfur cap for marine fuel) raised demand for low-sulfur blends and widened the hi-5 spread (HSFO vs VLSFO).
- Jet fuel tracks air travel; it trades at a premium/discount to diesel.

Fuel oil and residuals

- HSFO (3.5% S) demand comes from scrubber-fitted ships and Middle East power burn in summer.
- VLSFO (0.5% S) is the main marine fuel; Singapore and Fujairah are bunker hubs.

Key drivers

1. Refinery outages and run rates (EIA weekly utilisation).

2. Product inventories vs 5-year range (EIA, ARA Insights Global, Singapore IES).
3. Export arbitrage flows (US Gulf to LatAm/Europe; East of Suez to Europe).
4. Weather (heating degree days), freight (diesel trucking), hurricane hits on Gulf Coast refineries.
5. Structural capacity: closures in OECD vs new mega-refineries in Asia/Middle East/Africa (e.g., Dangote in Nigeria, ramping since 2024).

Desk notes

> Add your own observations here.

Natural Gas & LNG

Gas is regional by pipeline but increasingly global via LNG. The three benchmarks — **Henry Hub (US)**, **TTF (Europe)** and **JKM (North Asia)** — are linked by LNG shipping arbitrage.

Contract specifications

Contract	Venue / ticker	Size	Tick	Settlement
Henry Hub	NYMEX NG	10,000 MMBtu	$0.001 = $10	Physical at Henry Hub, Louisiana; expires 3 business days before the first calendar day of the delivery month
Dutch TTF	ICE Endex TFM	1 MW/hour for each hour of delivery month (e.g. 744 MWh in a 31-day month)	€0.005/MWh	Physical at TTF virtual hub
UK NBP	ICE M	1,000 therms/day	0.01 p/therm	Physical
JKM LNG	CME / ICE (Platts JKM)	10,000 MMBtu	$0.001	Cash-settled

Conversions: **1 MWh ≈ 3.412 MMBtu**; 1 therm = 0.1 MMBtu; 1 Bcf ≈ 1.037 million MMBtu (≈1 Tbtu); 1 tonne LNG ≈ 52 MMBtu ≈ 1,360 m³ gas.

US gas (Henry Hub)

- US produces ~100+ Bcf/d. Associated gas from the Permian (tied to oil drilling) and dry gas from Appalachia (Marcellus/Utica) and Haynesville (Louisiana; most price-elastic basin, close to LNG terminals).
- Demand: power (~35–40%), residential/commercial heating, industrial, LNG feedgas (rapid growth: Plaquemines, Corpus Christi Stage 3, Golden Pass), and pipeline exports to Mexico.
- **Storage:** injection season April–October, withdrawal November–March. The weekly EIA storage number vs consensus and the 5-year average is the key data point. End-October storage (~3.8–4.0 Tcf typical) and end-March storage are the critical balance points.
- **Weather is king:** heating/cooling degree days (HDD/CDD) from GFS and ECMWF model runs move prices intraday.
- **Permian bottlenecks:** Waha hub can price negative when pipelines are full; new takeaway pipes change this.
- **Widowmaker:** the March–April (H/J) spread — the bet on end-of-winter storage — is notoriously volatile.
- Data-center power demand is a newer structural growth narrative.

European gas (TTF)

- Post-2022 Europe replaced most Russian pipeline gas with LNG (US now the largest supplier) and Norwegian pipeline gas.
- **Storage regulation:** EU storage-filling targets (90% by 1 Nov historically, with added flexibility from 2025) drive summer buying; the summer–winter spread determines storage economics.
- Key supply risks: Norwegian maintenance/outages, LNG competition with Asia, end of Russian transit via Ukraine (expired end-2024), EU plan to phase out remaining Russian LNG.
- Demand: heating, power (gas sets the marginal power price in much of Europe; the clean spark spread vs coal and carbon), industry (demand destruction since 2022).

LNG

- Supply leaders: US, Qatar, Australia; growth wave 2025–2028 from the US and Qatar North Field expansion → widely expected to loosen the market.
- **Arbitrage:** US cargo goes to Asia vs Europe depending on JKM–TTF spread vs extra shipping cost (Panama/Suez/Cape routing).
- US export netback ≈ destination price − shipping − regas − liquefaction tolling fee (~115% × HH + fixed fee for US contracts). When netbacks fall below variable cost, US cargoes get cancelled (as in 2020).
- Oil-indexed long-term contracts in Asia (slope ~11–13% of Brent) compete with spot hub pricing.

Key drivers

1. Weather (temperature, wind/solar output, hydro levels).

2. Storage vs seasonal norms.
3. LNG outages (e.g., Freeport, Hammerfest, Gorgon) and new trains starting.
4. Geopolitics (Russia, Middle East — Qatar exports pass through Hormuz).
5. Coal-to-gas switching economics and carbon price (Europe).

Desk notes

> Add your own observations here.

Coal

Still the largest source of electricity generation globally, driven by China and India. Two families: **thermal (steam) coal** for power and **metallurgical (coking) coal** for steelmaking.

Benchmarks

Benchmark	Venue	Basis
API2	ICE (Rotterdam Coal)	CIF ARA, 6,000 kcal/kg NAR — Europe
Newcastle	ICE (globalCOAL NEWC)	FOB Newcastle, 6,000 kcal/kg NAR — Asia Pacific
API4	ICE (Richards Bay)	FOB South Africa, 6,000 kcal/kg NAR
Indonesian ICI	Argus/Coalindo	Lower-CV coal into China/India
Premium hard coking coal	SGX/CME (Platts PLV HCC FOB Australia)	Met coal benchmark
Zhengzhou thermal / Dalian coking coal & coke	ZCE / DCE	Chinese domestic

Contract size on ICE coal futures: **1,000 t/month**.

Fundamentals

- **Supply:** China (~50% of world output, mostly consumed domestically), India, Indonesia (largest thermal exporter), Australia (largest met coal exporter), Russia, South Africa, Colombia, US.
- **Demand:** Chinese and Indian power generation; China's hydro output and summer heat swing import needs. Europe is in structural decline but flexes with gas prices.
- Chinese domestic policy (safety inspections, output targets, import quotas, port restrictions) is the dominant price driver.
- Met coal follows Chinese and Indian steel output; Queensland weather (wet season Dec–Mar) disrupts supply.

Trading notes

- In Europe, coal vs gas is assessed through the **clean dark spread** (power − coal − carbon) vs the **clean spark spread** (power − gas − carbon).
- Newcastle–API2 spread reflects Pacific vs Atlantic balance and freight.
- Calorific-value adjustments matter: lower-CV coal trades at a discount roughly proportional to energy content but with additional quality penalties (ash, sulfur, moisture).

Desk notes

> Add your own observations here.

Electricity (Power)

Power cannot be stored economically at scale (batteries are changing this at the margin), so it is the most volatile commodity and priced by location and hour.

Key markets

Region	Venues / hubs	Notes
US	PJM West, ERCOT North/Houston, CAISO SP15/NP15, MISO, NYISO, ISO-NE	ICE and CME/Nodal futures; ERCOT is energy-only with scarcity pricing up to $5,000/MWh
Germany	EEX German baseload/peakload	Europe's benchmark; heavily renewable-influenced
France	EEX	Nuclear availability (EDF) drives winter risk
Nordics	Nasdaq / EEX (system price)	Hydro reservoir levels dominate
UK	ICE / EPEX	Gas-dominated marginal pricing

Concepts

- **Merit order:** generators ranked by marginal cost (renewables ≈ 0, nuclear, coal, CCGT gas, peakers). The marginal unit sets price.
- **Baseload vs peakload** products; on-peak/off-peak in the US (5×16, 2×16, 7×8).
- **Spark spread** = power − heat rate × gas price; **dark spread** = power − coal-equivalent. "Clean" versions subtract carbon cost.
- **Duck curve:** solar depresses midday prices, raising evening ramp prices; negative prices are increasingly common at midday.
- **Capacity markets** (PJM, UK) pay for availability, separate from energy prices.

Drivers

1. Weather (temperature, wind, solar irradiance, hydrology).
2. Fuel prices (gas, coal) and carbon (EUA, UKA, RGGI, California).
3. Plant outages (nuclear in France, thermal outages in ERCOT).
4. Interconnector flows and transmission congestion (basis between nodes and hubs).
5. Load growth from data centres, electrification and reshoring.

Desk notes

> Add your own observations here.

Precious Metals

Gold

Gold trades less like a consumable commodity and more like a monetary asset. Above-ground stocks (~215,000 t) dwarf annual mine supply (~3,600 t), so flows of investment and central-bank demand dominate price formation.

Contract specifications

Contract	Venue / ticker	Size	Tick	Notes
Gold	COMEX GC	100 troy oz	$0.10 = $10	Physical; 100-oz or 1-kg bars or 3 × 1-kg; active months G J M Q V Z
E-micro / Micro gold	COMEX MGC	10 oz	$0.10 = $1	
Loco London spot	OTC (LBMA)	400-oz good delivery bars	—	Unallocated vs allocated accounts; cleared via LPMCL
LBMA Gold Price	ICE Benchmark Administration	Auctions at 10:30 and 15:00 London	—	Used for contract settlement globally
Shanghai gold	SGE Au9999 / SHFE AU	1 kg	¥0.01/g	China premium/discount signals local demand
Gold ETFs	GLD, IAU, etc.	—	—	Holdings tracked daily as investment flow gauge

1 troy ounce = 31.1035 g; 1 tonne = 32,150.7 oz.

Supply

- Mine supply ~3,600 t/yr; China, Russia, Australia, Canada, US, Ghana are top producers. Mine supply is price-inelastic in the short run.
- Recycling ~1,100–1,400 t/yr and is the most price-elastic supply component (jewellery scrap surges on price spikes).
- All-in sustaining cost (AISC) for the industry gives a long-term cost reference, well below spot in recent years.

Demand

Segment	Driver
Jewellery (~45% of fabrication)	India and China; price-elastic; Indian wedding season and Diwali, Chinese New Year
Bar and coin	Retail investment; strong in Asia and Germany
ETFs	Western institutional flows; sensitive to real yields
Central banks	Reserve diversification away from USD since 2022 sanctions on Russia; >1,000 t/yr in 2022–2024 (PBoC, Poland, Turkey, India, etc.)
Technology	~7%; electronics

Price drivers

1. **US real interest rates** (10-year TIPS yield): historically the strongest inverse correlation. This relationship weakened from 2022 as central-bank and Asian buying dominated.
2. **US dollar:** generally inverse.
3. **Geopolitical risk / sanctions / de-dollarisation** — central bank buying.
4. Fiscal deficits, debt sustainability concerns and inflation expectations.
5. Fed policy path; gold tends to rally into cutting cycles.
6. Physical premiums in India/China and ETF flows as confirmation signals.

Market structure

- London OTC market is the hub; COMEX futures are the price-discovery venue in the US session.
- **EFP (COMEX vs London)** normally reflects financing and is small; in early 2025, tariff fears caused a large EFP blowout and massive metal flows from London to New York vaults.
- **Gold lease rates** (GOFO historically) show physical tightness.
- **Gold/silver ratio**, **gold/oil ratio** and **gold in other currencies** are standard cross-checks.

Trading notes

- Watch the 10-year TIPS yield, DXY, central bank purchase data (World Gold Council monthly), Swiss and Hong Kong customs export data to China/India.
- Gold often sells off in liquidity crises (margin calls) before rallying.
- Implied vol is typically low relative to other commodities; skew is call-skewed in strong bull markets.

Desk notes

Add your own observations here.

Silver

Hybrid metal: part monetary (follows gold), part industrial (over half of demand). More volatile than gold — typically 1.5–2× gold's volatility.

Contract specifications

Contract	Venue / ticker	Size	Tick	Notes
Silver	COMEX SI	5,000 troy oz	$0.005 = $25	Physical; active months H K N U Z
LBMA Silver Price	LBMA / CME Benchmark	Daily at 12:00 London	—	
Shanghai silver	SHFE AG	15 kg	¥1/kg	

London good delivery bars: ~1,000 oz.

Supply

- Mine supply ~820–830 Moz/yr; ~70% is a **by-product** of lead/zinc, copper and gold mining, so supply does not respond much to silver price.
- Top producers: Mexico, China, Peru, Chile, Bolivia, Poland, Russia.
- Recycling ~180–200 Moz.
- The market has been in **structural deficit** since 2021 (Silver Institute data), drawing down above-ground stocks.

Demand

- **Industrial (~55–60%)**: electronics, brazing, and especially **photovoltaics** (silver paste in solar cells); thrifting by solar manufacturers is the main counter-risk.
- Jewellery and silverware (India is the largest buyer).
- Investment: coins/bars, ETFs (SLV).

Price drivers

1. Gold direction and the gold/silver ratio (historically 50–90; extremes above 90 tend to mean-revert).
2. Solar installation growth (China), electronics cycle.
3. Indian import demand and duty changes.
4. Real rates and dollar (via gold).
5. London vault free-float tightness — lease rates spike when available metal is scarce.

Desk notes

> Add your own observations here.

Platinum Group Metals (Platinum, Palladium, Rhodium)

Small, concentrated, illiquid markets dominated by autocatalysts and by two producing countries.

Contract specifications

Contract	Venue / ticker	Size	Tick
Platinum	NYMEX PL	50 troy oz	$0.10 = $5
Palladium	NYMEX PA	100 troy oz	$0.50 = $50
London/Zurich spot	LPPM	OTC	—
Rhodium	OTC only (dealer quotes)	—	Very wide bid/ask

Supply

- **South Africa** (Bushveld Complex): ~70% of platinum, ~80% of rhodium supply. Risks: Eskom power cuts, labour strikes, deep-level mining costs, rand.
- **Russia** (Norilsk Nickel): ~40% of palladium supply (by-product of nickel).
- Zimbabwe, North America are smaller sources.
- Recycling of autocatalysts is significant (~25–30% of supply), and depends on scrap vehicle rates.

Demand

Metal	Main uses
Platinum	Diesel autocatalysts, heavy-duty vehicles, jewellery, glass/chemical industry, investment, hydrogen fuel cells/electrolysers (PEM)
Palladium	Gasoline autocatalysts (~80% of demand)
Rhodium	Gasoline autocatalysts (NOx reduction)

- **Battery EVs carry no autocatalyst** → long-term structural headwind for palladium/rhodium; hybrids carry more PGM loading than ICE.
- **Substitution:** high palladium prices drove platinum-for-palladium substitution in gasoline catalysts.
- Tightening emissions standards (Euro 7, China 6b) raise loadings.

Trading notes

- Platinum has been in deficit per World Platinum Investment Council data for several years; above-ground stocks cover the shortfall.
- Pt–Pd spread is a classic relative-value trade on substitution and EV penetration.
- Thin liquidity → sharp squeezes; watch NYMEX warehouse stocks and lease rates.
- Sanctions risk on Russian palladium is a perennial tail.

Desk notes

Add your own observations here.

Base Metals

The LME & Base Metals Market Structure

Most base metals price off the **London Metal Exchange (LME)**, with SHFE (Shanghai) and COMEX (copper) as important secondary venues.

LME mechanics

- **Prompt date structure:** daily prompts out to 3 months, weekly to 6 months, then monthly out to 123 months for some metals. The **3-month (3M)** is the liquid benchmark.
- **Cash–3s spread:** the key tightness indicator. Backwardation = prompt scarcity; contango = surplus.
- **Tom/next** and **cash/3s** spreads can spike when one entity holds dominant positions; the LME's **lending rules** cap backwardations for dominant holders.
- **Official prices** set in the second ring session (cash settlement for many physical contracts); **closing prices** at 17:00 London.
- Trading venues: Ring (open outcry, the only one remaining in Europe), LMEselect (electronic), inter-office telephone.
- Monthly average contracts (TAPOs/TAS) are used by physical traders who price on monthly averages.

Warehouses and stocks

- LME warrants are titles to metal in approved warehouses. **On-warrant** vs **cancelled warrants** (earmarked for withdrawal). Rising cancellations signal physical drawdown and often precede backwardations.
- Off-warrant stocks (reported monthly) and bonded Chinese stocks add a hidden layer.
- Since April 2024, Russian metal produced after the US/UK sanctions date cannot be delivered onto the LME.

Physical premiums

- Physical metal trades at LME + premium: e.g., Rotterdam duty-paid aluminium, US Midwest aluminium premium (CME), Yangshan copper premium (Chinese import demand), CIF Rotterdam nickel.
- Premiums often signal regional tightness before exchange prices do.

SHFE and arbitrage

- SHFE contracts are in yuan, VAT-inclusive (13%). The **import arbitrage** (SHFE vs LME adjusted for FX, VAT and duties) determines whether China pulls metal in.
- Chinese market hours overnight (in the Western time zone) often set the tone.

2022 nickel squeeze lesson

On 8 March 2022, LME nickel rose ~250% in two days due to a large short position (Tsingshan) and the Russia shock; the LME cancelled trades and suspended trading. It led to daily price limits and greater position transparency. **Lesson:** exchange rules can change mid-crisis; size positions for liquidity, not just price.

Desk notes

> Add your own observations here.

Copper

"Dr. Copper" — the bellwether of global industrial activity, now also the core energy-transition metal (grids, EVs, renewables, data centres).

Contract specifications

Contract	Venue / ticker	Size	Tick	Notes
LME Copper Grade A	LME CA	25 t	$0.50/t	Benchmark 3M
COMEX Copper	COMEX HG	25,000 lb	$0.0005/lb = $12.50	Months: all (active H K N U Z)
SHFE Copper	SHFE CU	5 t	¥10/t	VAT-inclusive

Conversion: **$/t ÷ 2,204.62 = $/lb**.

Supply

- Mined supply ~23 Mt/yr copper contained; refined ~27–28 Mt including secondary (scrap).
- Leading producers: **Chile** (~23–25%), **DRC** (fast-growing, now #2), **Peru**, **China**, US, Russia, Indonesia, Australia, Zambia.
- **China** is ~45–50% of refined (smelting) output: it imports concentrate and refines.
- **Treatment and refining charges (TC/RCs):** fees smelters charge miners. Low or negative TCs (as in 2024–2025) mean concentrate is scarce relative to smelting capacity — a mine supply tightness signal.
- Long lead times: a new mine takes 10–15+ years from discovery to production; grades are declining (Chile).
- Disruption allowance: analysts assume 4–6% of mine supply is lost annually to strikes, accidents, weather, political events (e.g., Cobre Panamá closure 2023, Grasberg mudslide 2025, Kamoa-Kakula seismic event 2025).

Demand

Sector	Share (approx.)
Construction (wiring, plumbing)	~28%
Electrical grid / power	~28%
Consumer & industrial equipment	~20%
Transport (incl. EVs — ~80 kg per BEV vs ~20 kg ICE)	~12%

- **China consumes ~55% of refined copper.** Grid investment (State Grid), property, and manufacturing exports matter.
- Energy-transition and AI data-centre demand provide the structural bull case; property weakness in China the bear case.

Price drivers

1. Chinese demand: PMIs, credit (TSF), property, State Grid spend, Yangshan premium, SHFE stocks.
2. Mine disruptions and TC/RC levels.
3. USD and global growth expectations.
4. Inventory location across LME/COMEX/SHFE/bonded.
5. **US tariff policy** — the COMEX–LME arbitrage blew out in 2025 on Section 232 tariff expectations, pulling large stocks into the US.
6. Scrap availability (price-elastic supply).

Trading notes

- Monitor COMEX–LME spread (tariff/arbitrage), LME cash-3s, cancelled warrants, Chinese import arbitrage and Yangshan premium.
- Copper/gold ratio tracks US 10-year yields and growth sentiment.
- Seasonality: Chinese demand strong in Q2 ("golden March/April") and soft around Lunar New Year.

Desk notes

> Add your own observations here.

Aluminium

The most consumed non-ferrous metal (~70 Mt/yr primary). Production is an energy-intensive process — "congealed electricity" — so power costs drive the cost curve.

Contract specifications

Contract	Venue	Size	Notes
LME Primary Aluminium	LME AH	25 t	P1020A, min 99.7% Al
SHFE Aluminium	SHFE AL	5 t	
US Midwest premium	CME AUP	25 t	$/lb, Platts
Rotterdam duty-paid premium	CME	25 t	$/t
Alumina	CME (Platts FOB Australia)	1,000 t (cash-settled)	

Value chain

- **Bauxite → alumina → aluminium**: ~4–5 t bauxite → ~2 t alumina → 1 t aluminium.
- Smelting uses ~13–15 MWh per tonne; electricity is ~35–40% of cost.
- Bauxite: Guinea, Australia, China; alumina: China, Australia; aluminium: **China ~60% of world output**.

Supply dynamics

- China imposed a **45 Mt capacity cap** — growth is shifting to Indonesia, India, the Middle East.
- Yunnan hydro shortages periodically curtail Chinese output in the dry season.
- European smelters curtailed in 2022 on power costs; restarts are slow.
- Russian supply (Rusal) faces sanctions; can't be delivered on LME (post-April 2024 production).

Demand

- Transport (lightweighting, EVs), construction, packaging (cans), electrical (grid cables), consumer durables.
- Recycling (secondary aluminium) uses ~5% of the energy of primary.

Price drivers

1. Energy costs (Chinese coal, European power).
2. Chinese production vs capacity cap; exports of semi-finished products.
3. Trade policy: US Section 232 tariffs, EU CBAM (carbon border adjustment, definitive phase from 2026), anti-dumping duties.
4. Alumina price spikes (2024 spike on bauxite and refinery outages).
5. Regional premiums — the **US Midwest premium** rises with tariffs.

Desk notes

> Add your own observations here.

Zinc & Lead

Often mined together (and with silver) from the same ore bodies.

Contract specifications

Contract	Venue	Size	Notes
LME Special High Grade Zinc	LME ZS	25 t	99.995% Zn
LME Lead	LME PB	25 t	
SHFE Zinc / Lead	SHFE ZN / PB	5 t	

Zinc

- ~13–14 Mt/yr. Main use: **galvanising steel (~50–60%)**, die-casting alloys, brass, chemicals.
- Demand tied to construction, infrastructure, autos — very China-centric.
- Mine supply: China, Peru, Australia, India, US, Mexico. Smelting dominated by China.
- **Zinc TC/RCs** are a key signal (annual benchmark negotiated between miners and smelters; spot TCs move monthly).
- European smelters are sensitive to energy prices (Nyrstar, Glencore curtailments 2022).

Lead

- ~12–13 Mt/yr. **Lead-acid batteries ~80%** of demand (automotive starter batteries, backup power, e-bikes in Asia).
- Over 60% of supply is **secondary** (recycled batteries) — making lead the most recycled metal. Supply is therefore stable and price-insensitive.
- Replacement-battery demand is weather-sensitive (cold winters, hot summers kill batteries).
- Long-term headwind from lithium-ion substitution, but ICE and EV 12V systems still need lead-acid.

Desk notes

Add your own observations here.

Nickel

A market transformed by Indonesia, which grew from a small producer to more than half of global supply in under a decade.

Contract specifications

Contract	Venue	Size	Notes
LME Nickel	LME NI	6 t	Class 1 (≥99.8% Ni); daily price limits since 2022
SHFE Nickel	SHFE NI	1 t	

Product types

- **Class 1** (≥99.8%): cathodes, briquettes, powder — LME deliverable; used in batteries (via sulphate) and alloys.
- **Class 2** (<99.8%): **nickel pig iron (NPI)** and ferronickel — used in stainless steel.
- **Intermediates:** mixed hydroxide precipitate (MHP), nickel matte — feed for battery sulphate.

Supply

- **Indonesia (>50%)**: built on Chinese-financed NPI and HPAL (high-pressure acid leach) plants after the 2020 nickel ore export ban. Government RKAB mining quotas are now a key policy lever.
- Philippines (ore exports), Russia (Norilsk — Class 1), Canada, Australia (many mines closed in 2024 due to low prices), New Caledonia (crisis, unrest).

Demand

- **Stainless steel ~65–70%** (China dominant).
- **Batteries ~15%** and growing (NMC chemistries) — but LFP batteries (no nickel) have taken share.
- Alloys, plating.

Price drivers

1. Indonesian supply policy (quotas, royalties, export restrictions).
2. Chinese stainless output and margins.
3. EV battery chemistry mix (NMC vs LFP).
4. LME stocks: Class 1 from Indonesia/China now registered as LME brands, raising deliverable stocks.

Trading notes

- The LME price reflects only Class 1, yet most production is Class 2 → disconnect between LME and the bulk of the physical market; watch NPI prices and premiums.
- The 2022 short squeeze reduced LME liquidity for years; daily limits apply.

Desk notes

> Add your own observations here.

Tin

The smallest and most volatile LME base metal (~380 kt/yr), with highly concentrated and fragile supply.

Contract specifications

Contract	Venue	Size
LME Tin	LME SN	5 t
SHFE Tin	SHFE SN	1 t

Fundamentals

- **Demand:** solder (~50%) — electronics and semiconductors; tinplate (packaging); chemicals; lead-acid battery additives. Semiconductor/AI cycle is the key demand driver.
- **Supply:** China, Indonesia (export licence/policy changes, crackdown on illegal mining), Myanmar (Wa State — mining suspended from 2023, slow restart), Peru, DRC (Bisie mine — conflict risk), Bolivia, Brazil.
- Very low LME stocks relative to consumption → violent squeezes and backwardations.

Trading notes

- Watch Indonesian export data (monthly), Myanmar Wa State news, Chinese smelter feedstock (concentrate imports), Philadelphia Semiconductor Index (SOX).
- Liquidity is thin; size positions conservatively.

Desk notes

Add your own observations here.

Bulks, Battery Metals & Uranium

Iron Ore & Steel

The largest seaborne dry-bulk trade (~1.6 Bt/yr seaborne iron ore). Pricing is effectively a bet on Chinese steel production.

Contract specifications

Contract	Venue	Size	Basis
Iron Ore 62% Fe	SGX FEF	100 t	Cash-settled vs Platts IODEX 62% Fe CFR Qingdao (monthly average)
Iron Ore 62% Fe	CME TIO	500 t	Cash-settled
Iron Ore	DCE I	100 t	Physical, yuan, port-stock basis
Hot-Rolled Coil (US Midwest)	CME HRC	20 short tons	CRU index
Rebar / HRC	SHFE RB / HC	10 t	Physical, China
Coking coal / coke	DCE JM / J	60 t / 100 t	Physical

Supply

- **Australia (~55% of seaborne)**: Rio Tinto, BHP, Fortescue in the Pilbara — lowest-cost producers.
- **Brazil (~25%)**: Vale; higher-grade (65% Fe) Carajás ore; rainy season (Dec–Apr) cuts shipments.
- **Simandou (Guinea)**: giant high-grade project, first shipments from late 2025 — a major medium-term supply increase.
- Others: South Africa, India, Canada, Ukraine.
- Quarterly production reports from the majors are important supply signals; Q1 is weather-impacted (Pilbara cyclones).

Demand

- **China makes ~53–55% of world crude steel (~1 Bt/yr)** and imports ~70% of seaborne ore.
- Blast furnace (BF-BOF) steelmaking uses ore + coking coal; electric arc furnaces (EAF) use scrap and DRI. India is the main growth market.
- Chinese steel demand: property (structural decline since 2021), infrastructure, manufacturing, and **steel exports** (record levels, triggering trade defense worldwide).
- Production controls/curbs (environmental, output caps) can suddenly cut ore demand.

Grade and quality

- **62% Fe** is the benchmark; 65% Fe high-grade (Brazil, Canada) earns a premium that widens when steel margins are high (mills want productivity) and narrows when mills cut costs.
- 58% Fe low-grade discounts widen when margins are good.
- Penalties for alumina, silica, phosphorus.

Price drivers and indicators

1. Chinese steel margins (rebar/HRC minus ore and coke costs) → mill demand.
2. Chinese port stocks (Mysteel ~45 ports) and mill inventories (days of use).
3. Property starts, local government special bond issuance, infrastructure stimulus.
4. Steel export volumes and trade barriers.
5. Supply: Australian/Brazilian shipment data (weekly), Simandou ramp-up.
6. Freight: Capesize rates (Brazil–China C3, West Australia–China C5).

Desk notes

Add your own observations here.

Battery Metals: Lithium, Cobalt, Graphite, Manganese

Young, opaque markets with limited exchange liquidity. Prices are set by price-reporting agencies (Fastmarkets, Benchmark, SMM) and Chinese futures.

Venues

Contract	Venue	Size	Basis
Lithium carbonate	GFEX (Guangzhou) LC	1 t	Physical, battery grade, China
Lithium hydroxide CIF CJK	CME	1 t	Cash-settled vs Fastmarkets
Lithium carbonate	CME / LME	1 t	Cash-settled
Spodumene (SC6)	SGX / CME	1 t	Cash-settled
Cobalt metal	CME / LME	1 t	Fastmarkets standard grade
Cobalt hydroxide	CME	1 t	Fastmarkets payable

Lithium

- **Supply:** hard rock (spodumene) — Australia (Greenbushes, Pilgangoora); brines — Chile (SQM, Albemarle), Argentina (rapid growth); **lepidolite** in Jiangxi, China (high-cost marginal supply); Zimbabwe; Africa growing. China dominates chemical conversion (~65%+).
- **Demand:** EV batteries (~75%+), energy storage (ESS — fastest-growing), consumer electronics, ceramics/glass.
- **Price cycle:** extreme boom in 2021–2022 followed by a collapse of more than 80% by 2024–2025 on supply overbuild. Mine closures and Chinese lepidolite permit issues (e.g., CATL Jianxiawo 2025) drive rallies.
- Units: LCE (lithium carbonate equivalent); spodumene concentrate ~6% Li2O (SC6). Rule of thumb: ~7.5–8 t SC6 per t LCE.

Cobalt

- **DRC supplies ~75%** of mined cobalt, mostly as a copper by-product (CMOC, Glencore). China refines ~75%+.
- Demand: batteries (NMC, NCA), superalloys (aerospace), hard metals.
- **DRC export ban (Feb 2025)** then quota system — a supply-policy-driven market. LFP batteries (cobalt-free) cap long-term demand growth.
- Artisanal mining and ESG/child-labour concerns affect Western buyer sourcing.

Graphite and manganese

- Graphite (anode material): China dominates natural and synthetic graphite processing; Chinese export controls (since Dec 2023) are a key risk.
- Manganese: steel (~90%) and LMFP/NMC battery cathodes; South Africa, Gabon, Australia.

Trading notes

- Watch Chinese EV sales (CPCA weekly/monthly), battery production (CABIA), ESS tenders, GFEX open interest, Australian miner quarterlies, Chilean/Argentine export data.
- Policy (China export controls, US IRA/FEOC rules, DRC quotas) can override fundamentals.

Desk notes

Add your own observations here.

Uranium

Mostly traded via long-term contracts between utilities and producers; the spot market is thin and dominated by traders, financial vehicles (Sprott Physical Uranium Trust, Yellow Cake plc) and producers.

Pricing

Market	Basis
Spot U3O8	UxC / TradeTech; $/lb
Long-term price	Monthly indicator; more stable, base-escalated contracts
CME UxC U3O8 futures	250 lb, cash-settled — limited liquidity
Conversion (UF6) and enrichment (SWU)	Separate fuel-cycle prices, both tight since 2022

Fuel cycle

Mining (U3O8 "yellowcake") → conversion (UF6) → enrichment (SWU) → fuel fabrication → reactor. Each step is concentrated: **Russia (Rosatom) holds ~40% of global enrichment capacity**; the US banned Russian enriched uranium imports from 2024 (waivers until 2028).

Supply

- **Kazakhstan (Kazatomprom) ~40%** of mined supply (in-situ recovery); constrained by sulphuric acid availability and a higher mineral extraction tax from 2025.
- **Canada (Cameco, Athabasca Basin)**: McArthur River/Key Lake; Cigar Lake.
- Namibia, Australia, Uzbekistan, Niger (coup 2023, Orano lost control of Somair).
- Secondary supply (stockpiles, underfeeding at enrichment plants) bridges the primary deficit.

Demand

- ~440 operating reactors consuming ~180–190 Mlb U3O8/yr. Growth: China (largest build pipeline), India, reactor life extensions, restarts (Japan), and new interest from tech companies for data centres; SMRs are a longer-term option.
- Utility contracting cycles: under-contracting since the 2010s creates periodic term-buying waves.

Desk notes

> Add your own observations here.

Grains & Oilseeds

Corn (Maize)

The largest grain crop by volume (~1.2 Bt/yr globally). The US is the largest producer and exporter, but Brazil's second-crop (safrinha) corn has made it a year-round competitor.

Contract specifications

Contract	Venue / ticker	Size	Tick	Months
Corn	CBOT ZC	5,000 bu	¼¢/bu = $12.50	H K N U Z
Mini corn	CBOT XC	1,000 bu	⅛¢	Same
Euronext corn	Euronext EMA	50 t	€0.25/t	H M Q X
Dalian corn	DCE C	10 t	¥1/t	
B3 corn	B3 CCM	450 sacks (60 kg)	R$0.01	Brazil

Conversions: **1 bu corn = 56 lb = 25.4 kg; 39.37 bu/t**. Price limits apply (reset semi-annually).

Balance sheet

- **US:** ~90 million acres planted, trend yield ~180+ bu/acre; production ~15+ billion bu.
- Use: **feed & residual (~38–40%)**, **ethanol (~35%)**, exports (~15–20%), food/seed/industrial.
- **Stocks-to-use** ratio is the key price metric; below ~10% is tight, above ~15% is comfortable.
- Major exporters: US, Brazil, Argentina, Ukraine. Importers: Mexico (#1 for US corn), Japan, China (volatile), South Korea, EU, Colombia, Vietnam.

Crop calendar

Region	Planting	Critical stage	Harvest
US	Apr–May	**Pollination: July**	Sep–Nov
Brazil 1st crop	Sep–Nov	Dec–Jan	Feb–Apr
Brazil safrinha (~75% of Brazil output)	Jan–Mar (after soy)	Apr–May (rains end)	Jun–Aug
Argentina	Sep–Dec	Jan–Feb	Mar–Jul
Ukraine	Apr–May	Jul	Sep–Nov

Price drivers

1. **US weather** June–August (ridge-of-high-pressure patterns, heat during pollination). The "weather premium" is usually built into Dec futures by early summer and removed if conditions are good ("buy the rumour, sell by 4th of July").
2. USDA reports: March Prospective Plantings, June 30 Acreage & Grain Stocks, monthly WASDE, August crop production (first survey-based yield), January Final.
3. Ethanol: EIA weekly production, gasoline demand, ethanol margins, RFS policy (RVOs), E15 waivers.
4. Export competitiveness vs Brazil (FOB Paranaguá vs FOB Gulf) and the BRL.
5. Chinese buying.
6. Corn–soy acreage battle: the **Nov soy / Dec corn price ratio** (above ~2.3–2.5 favours soy; below favours corn).
7. Input costs (nitrogen fertiliser, natural gas).

Spreads and trading notes

- **Old-crop/new-crop (July/Dec)** spread reflects ending-stocks tightness.
- Corn–wheat spread: wheat feeds when it is cheap relative to corn.
- Basis: Gulf CIF barge, interior elevators; Mississippi River low water (autumn) widens basis and raises barge freight.
- Seasonal: harvest lows Oct–early Nov are common; post-harvest carry rally.

Desk notes

Add your own observations here.

Wheat

The most widely traded food grain (~800 Mt produced; ~210–220 Mt traded). Unlike corn, global supply is diversified across both hemispheres, and **Black Sea exporters** dominate.

Contract specifications

Contract	Venue / ticker	Class	Size	Tick	Months
Chicago SRW	CBOT ZW	Soft red winter	5,000 bu	¼¢ = $12.50	H K N U Z
KC HRW	CBOT KE	Hard red winter	5,000 bu	¼¢ = $12.50	H K N U Z
Minneapolis HRS	MIAX (ex-MGEX) MWE	Hard red spring	5,000 bu	¼¢	H K N U Z
Milling wheat No.2	Euronext EBM (Matif)	EU milling	50 t	€0.25/t	Sep, Dec, Mar, May
Black Sea wheat	CME (Platts)	12.5% protein FOB	50 t	Cash-settled	Monthly

Conversions: **1 bu wheat = 60 lb; 36.74 bu/t**.

Classes

- **HRW**: bread wheat, Great Plains (Kansas, Oklahoma, Texas); winter crop planted Sep–Oct, harvested Jun–Jul.
- **SRW**: crackers, cookies; eastern Corn Belt; most liquid but lowest-protein contract.
- **HRS**: high protein (13–14%); Northern Plains and Canada; planted spring, harvested Aug–Sep.
- **Durum**: pasta; Canada, EU, North Africa.
- Protein spreads (KC–Chicago, Minneapolis–KC) widen in years of low protein supply.

Global supply

Exporter	Approx. annual exports	Notes
Russia	40–50 Mt	Largest exporter; export taxes/quotas and unofficial floor prices; Novorossiysk
EU	30–35 Mt	France, Romania, Germany, Poland; Rouen/Constanta
Canada	20–25 Mt	HRS, durum
Australia	20–30 Mt	Harvest Nov–Jan; drought-volatile (El Niño)
US	20–23 Mt	
Ukraine	10–20 Mt	War-affected; Black Sea corridor
Argentina	8–15 Mt	Harvest Dec–Jan
Kazakhstan	~8–10 Mt	

Big importers: Egypt (GASC/Mostakbal Misr tenders), Indonesia, Turkey, Algeria (OAIC), Saudi Arabia (GFSA), China, Philippines, Nigeria, Bangladesh. **Tender results** are price-discovery events for Black Sea/EU FOB values.

Price drivers

1. Black Sea crop size, Russian export policy and geopolitics.
2. Northern Hemisphere weather: US Plains drought, EU/Russia spring dryness, winterkill, harvest rains (quality).
3. Southern Hemisphere: Australia and Argentina (ENSO-driven).
4. Global stocks — note China holds over half of world wheat stocks, which are not available to the market; use "major exporters' stocks-to-use".
5. Corn–wheat spread (feed substitution).
6. USD and EUR/USD (US vs EU competitiveness).

Trading notes

- US wheat often trades as the "market of last resort" — the US gains exports only when Black Sea supplies run short.
- Managed money is frequently structurally short Chicago wheat; short-covering rallies on weather/geopolitical headlines are violent.
- Euronext–CBOT spread reflects EU vs US competitiveness after FX.

Desk notes

Add your own observations here.

Soybean Complex (Soybeans, Meal, Oil)

Soybeans are crushed into **meal** (protein for animal feed) and **oil** (food and biofuels). The complex is driven by Brazil supply, Chinese import demand, and increasingly US biofuel policy.

Contract specifications

Contract	Venue / ticker	Size	Tick	Months
Soybeans	CBOT ZS	5,000 bu	¼¢/bu = $12.50	F H K N Q U X
Soybean meal	CBOT ZM	100 short tons	$0.10/st = $10	F H K N Q U V Z
Soybean oil	CBOT ZL	60,000 lb	0.01¢/lb = $6	F H K N Q U V Z
Dalian soybean No.1/No.2, meal, oil	DCE A / B / M / Y	10 t	¥1/t	

1 bu soybeans = 60 lb. A bushel crushed yields ~**11 lb oil** and ~**44 lb (0.022 st) meal** (48% protein), plus hulls.

Board crush margin

Crush ($/bu) = (Meal $/st × 0.022) + (Oil ¢/lb × 0.11) − Beans $/bu. Oil share of crush value (oil value ÷ total product value) has risen structurally from ~30–35% to ~40–50% due to renewable diesel demand.

Supply

Producer	Share	Calendar
Brazil	~40%, largest exporter (~100+ Mt exports)	Plant Sep–Dec, harvest Jan–Apr (Mato Grosso first)
United States	~28%	Plant May–Jun, pod-fill **August**, harvest Sep–Nov
Argentina	~12–14%	Largest **meal and oil** exporter (crushes domestically); export taxes (retenciones) a key policy variable
Paraguay, China, India, Canada	smaller	

Demand

- **China imports ~100–110 Mt of soybeans/yr** (~60% of world trade) to crush for hog and poultry feed. US–China trade relations shift flows between US and Brazil (in tariff disputes, China buys almost exclusively from Brazil).
- Meal demand follows livestock numbers (China hogs, EU, Southeast Asia aquaculture).
- **Soybean oil** demand: cooking, food, and **biofuels** — US renewable diesel/SAF under RFS (RVOs), 45Z clean fuel credit, California LCFS; Brazil and Argentina/Indonesia biodiesel mandates.

Price drivers

1. Brazilian crop size (CONAB, weather in Mato Grosso, Rio Grande do Sul) and BRL.
2. US August weather; USDA June Acreage and August/September yields.
3. Chinese purchases (USDA flash sales) and US–China trade policy.
4. US biofuel policy (EPA RVO announcements, small refinery exemptions) → soyoil.
5. Argentine export tax changes and farmer selling.
6. Palm oil (substitute veg oil) and crude oil/heating oil (BOHO spread = soyoil vs heating oil — biofuel economics).

Trading notes

- **Oil share** and **meal–oil spread** trades reflect biofuel demand vs feed demand.
- Brazil premiums (Paranaguá FOB basis) vs US Gulf basis show export competition.
- Jul/Nov old-crop/new-crop and Nov soy/Dec corn ratio for acreage.

Desk notes

Add your own observations here.

Vegetable Oils: Palm, Rapeseed/Canola, Sunflower

Veg oils are a single substitutable pool (~220+ Mt/yr): palm, soy, rapeseed and sunflower oils compete on price, with biofuel mandates adding energy linkage.

Contract specifications

Contract	Venue	Size	Currency
Crude Palm Oil (FCPO)	Bursa Malaysia	25 t	MYR/t — 3rd month is benchmark
Palm olein	DCE P	10 t	CNY
Canola	ICE Futures US (RS)	20 t	CAD/t; months F H K N X
Rapeseed	Euronext ECO	50 t	EUR/t; months Feb, May, Aug, Nov
Soybean oil	CBOT ZL	60,000 lb	US¢/lb

Palm oil

- **Indonesia (~58%) and Malaysia (~25%)** produce ~85% of world palm oil. Cheapest veg oil by yield per hectare.
- **Seasonality:** output rises Mar–Oct, peaks Sep–Oct, troughs Jan–Feb. A lagged response to rainfall; El Niño droughts cut yields 6–12 months later.
- **Indonesian policy is the dominant driver:** biodiesel mandate (B40 implemented 2025, B50 planned), export levies and DMO rules. Higher mandates remove exportable supply.
- Aging trees and slow replanting limit growth; labour shortages in Malaysia.
- Data: **MPOB** monthly (stocks, production, exports); cargo surveyors (ITS, AmSpec) every 5/10 days.
- Demand: India (largest importer, sensitive to import duty changes), China, EU (EUDR deforestation regulation), Pakistan.

Rapeseed / canola

- Canada (canola), EU, Australia, Ukraine. EU uses rapeseed oil mainly for biodiesel.
- Canola crush in Canada is expanding for renewable diesel; China–Canada trade tariffs (2025) hit canola meal/oil.

Sunflower oil

- **Ukraine and Russia ~55–60% of world production and ~75%+ of exports**; war risk premium.

Key spreads

- **POGO** (palm oil – gasoil): biodiesel blending economics.
- **Soy–palm spread (SBO–CPO)**: when palm discount narrows too much, buyers (India) switch to soy oil.
- Rapeseed–soybean oil spread in the EU.

Desk notes

Add your own observations here.

Rice, Oats & Minor Grains

Rice

- ~520 Mt (milled) produced; only ~10% traded — a thin international market with heavy government involvement.
- Exporters: **India (~40% of trade)**, Thailand, Vietnam, Pakistan, US, Myanmar. India's export bans/duties (2022–2024, then lifted 2024–25) move world prices sharply.
- Benchmarks: Thai 5% broken FOB, Vietnam 5% broken, Indian parboiled.
- **CBOT Rough Rice (ZR):** 2,000 cwt (hundredweight), $0.005/cwt = $10; US long-grain (Arkansas). Illiquid.
- Drivers: monsoon (India), El Niño, government stockholding, Philippines/Indonesia import tenders.

Oats

- **CBOT Oats (ZO):** 5,000 bu, ¼¢; 1 bu = 32 lb. Supply mainly from Canada (Prairies) and Nordics; demand from food (oat milk, cereals) and feed/horse. Very thin, prone to squeezes.

Barley and sorghum

- Barley: feed and malting; EU, Russia, Australia, Canada; China is a major importer. Traded mostly physically.
- Sorghum: US and Argentina export, China the main buyer — closely tied to US–China trade and corn.

Desk notes

> Add your own observations here.

Softs

Coffee

Two distinct species with separate futures: **Arabica** (higher quality, ~55–60% of output) and **Robusta** (hardier, higher caffeine, used in instant coffee and blends). Global production ~175 million 60-kg bags.

Contract specifications

Contract	Venue / ticker	Size	Tick	Months
Coffee "C" (Arabica)	ICE US KC	37,500 lb	0.05¢/lb = $18.75	H K N U Z
Robusta	ICE Europe RC	10 t	$1/t = $10	F H K N U X
Brazil Arabica	B3 ICF	100 bags	US$	

Conversion: **Robusta $/t ÷ 22.046 = ¢/lb**. The **Arabica–Robusta arbitrage** (KC ¢/lb − RC converted to ¢/lb) shows the relative premium.

Supply

Origin	Type	Calendar / notes
Brazil (~35–40%)	Arabica + Robusta (conilon)	Harvest May–Sep; **frost risk Jun–Aug** (Minas Gerais); flowering Sep–Oct needs rain; Arabica on-/off-year biennial cycle
Vietnam (~17%)	Robusta	Harvest Oct–Jan; Central Highlands drought risk
Colombia	Washed Arabica	Two harvests; La Niña rains cut output
Indonesia	Robusta (Sumatra, Lampung)	Harvest Apr–Jul
Ethiopia, Honduras, Uganda, India, Peru, Central America	various	

- Arabica trees take ~3–4 years to mature → slow supply response.
- **ICE certified stocks** (graded, exchange-deliverable) are a closely watched tightness indicator.

Demand

- ~170+ Mbags/yr; EU, US, Brazil (big domestic consumer), Japan; fastest growth in Asia (China, Indonesia).
- Demand is fairly price-inelastic at retail; roasters change blends (more robusta) when arabica is expensive.
- **EUDR** (EU deforestation regulation) compliance costs and timing can create two-tier pricing.

Price drivers

1. Brazilian weather (frost, drought) and crop estimates (CONAB, USDA attaché, private analysts).
2. **BRL**: weaker real encourages Brazilian farmer selling in USD terms (bearish).
3. Vietnamese weather and farmer hoarding.
4. ICE certified stocks; freight/container disruptions.
5. US tariffs on origin countries (2025 tariffs on Brazil caused trade-flow and price dislocations).
6. Speculative positioning; the 2024–2025 rally took arabica to record highs on consecutive Brazil/Vietnam shortfalls.

Trading notes

- Differentials (physical premium over futures for a specific origin) matter as much as futures for physical traders.
- Backwardation signals tight nearby availability; certified stock trends confirm.
- Options skew is call-heavy into the Brazilian winter.

Desk notes

Add your own observations here.

Cocoa

The most concentrated soft: **Côte d'Ivoire and Ghana produce ~55–60% of world cocoa**. A structural shock (disease, weather, underinvestment) took prices to record highs above $12,000/t in 2024 before demand destruction set in.

Contract specifications

Contract	Venue / ticker	Size	Tick	Months
Cocoa (NY)	ICE US CC	10 t	$1/t = $10	H K N U Z
London Cocoa	ICE Europe C	10 t	£1/t = £10	H K N U Z

The **London–NY arbitrage** (adjusted for GBP/USD) reflects European vs North American availability and the pound.

Supply

- West Africa ~70%: Côte d'Ivoire (~2 Mt in normal years), Ghana, Cameroon, Nigeria; plus Ecuador (fast-growing), Brazil, Indonesia.
- **Main crop** Oct–Mar; **mid crop** Apr–Sep (smaller).
- Structural problems: **swollen shoot virus**, aging trees, illegal gold mining (galamsey) in Ghana, low farmgate prices historically, black pod disease in wet seasons, Harmattan dry winds (Dec–Feb).
- Government marketing boards fix farmgate prices and **forward-sell** crops (CCC in Côte d'Ivoire, COCOBOD in Ghana). Their forward sales and any defaults on contracts are important signals.
- **Living Income Differential (LID):** $400/t premium on Ivorian/Ghanaian cocoa since 2020.

Demand

- Grindings (processing beans into butter, powder and liquor) are the demand proxy: quarterly **European (ECA), North American (NCA), Asian grindings** data.
- Price-elastic at very high prices: manufacturers reformulate (less cocoa content, smaller sizes).
- Grinders: Barry Callebaut, Cargill, Olam (ofi); chocolate: Mars, Mondelez, Hershey, Nestlé, Ferrero.

Price drivers

1. Weekly **Ivorian port arrivals** vs last year.
2. West African weather: rainfall, soil moisture, Harmattan intensity.
3. Quarterly grind data.
4. ICE certified stocks (multi-decade lows in 2024).
5. ICCO balance estimates.
6. Farmgate price decisions and smuggling flows between countries.

Trading notes

- Liquidity fell during 2024 as margin requirements surged; this amplified volatility and moved hedging to OTC.
- Bean quality (bean count, moisture) affects certification eligibility.

Desk notes

Add your own observations here.

Sugar

~180 Mt produced annually; ~80% from sugarcane, ~20% from sugar beet. The market is shaped by Brazil's choice between making sugar or ethanol, and by Indian policy.

Contract specifications

Contract	Venue / ticker	Product	Size	Tick	Months
Sugar No.11	ICE US SB	Raw cane sugar, FOB origin	112,000 lb (~50.8 t)	0.01¢/lb = $11.20	H K N V
White Sugar No.5	ICE Europe W	Refined white sugar, FOB	50 t	$0.10/t	H K Q V Z
Sugar No.16	ICE US	US domestic	112,000 lb		Protected market

White premium = No.5 ($/t) − No.11 (¢/lb × 22.0462). A premium above ~$100–110/t typically makes refining raw sugar profitable for standalone refiners.

Supply

Producer	Notes
Brazil (~40%+ of exports)	Centre-South crushes Apr–Nov; mills switch cane between sugar and ethanol based on relative returns (**sugar mix** — reported by **UNICA** bi-weekly)
India	#2 producer; season Oct–Sep; monsoon-dependent; government export quotas and ethanol diversion
Thailand	Major raw exporter; Nov–Apr crush
EU (beet), China, US, Australia, Mexico, Pakistan	

Ethanol parity

- Brazilian mills compare sugar prices with **hydrous ethanol** prices (set by Petrobras gasoline pricing, BRL, and the flex-fuel fleet). Sugar must exceed ethanol parity for mills to maximise sugar.
- Higher crude/gasoline prices → more cane to ethanol → bullish sugar. Brazil's ethanol blend in gasoline (E30 from 2025) adds demand.

Price drivers

1. Brazilian CS weather (rain, frost, fire risk in dry season) and cane yield (ATR, TRS per tonne).
2. UNICA sugar mix and crush pace.
3. Indian monsoon, cane area, export policy and ethanol diversion.
4. Thai production.
5. BRL, crude oil.
6. Import demand (China, Indonesia, Middle East refiners).

Trading notes

- Spreads: March/May and the **White premium**. V (Oct) expiry delivery size signals physical demand.
- Raw sugar delivery tonnage at expiry and the receiver's identity are watched closely.

Desk notes

Add your own observations here.

Cotton

The main natural fibre (~115–120 million 480-lb bales/yr); competes with polyester (an oil derivative).

Contract specifications

Contract	Venue / ticker	Size	Tick	Months
Cotton No.2	ICE US CT	50,000 lb (~100 bales)	0.01¢/lb = $5	H K N V Z
Zhengzhou cotton	ZCE CF	5 t	¥5/t	
Cotlook A Index	Physical benchmark	—	—	CFR Far East average

Supply

- China (Xinjiang), India, **Brazil (now #1 exporter)**, US (#2 exporter; Texas dryland is weather-critical), Pakistan, Australia, Uzbekistan, Turkey.
- US calendar: plant Apr–Jun, boll set Jul–Aug, harvest Sep–Dec. Hurricanes affect Southeast/Delta crops.
- **Texas West Plains abandonment** (dryland drought) can swing the US crop by millions of bales.

Demand

- Mill use: China, India, Pakistan, Bangladesh, Vietnam, Turkey. Demand follows apparel retail and the macro cycle.
- **US Uyghur Forced Labor Prevention Act** bans Xinjiang cotton in US supply chains — reshapes trade flows.
- Polyester price (via crude/PTA) affects fibre substitution.

Price drivers

1. US weather and crop conditions; USDA weekly export sales (key every Thursday).
2. Chinese reserve policy and import quotas.
3. Global textile demand, retail inventories.
4. **On-call sales/purchases** report (CFTC weekly): unfixed mill purchases are latent futures buying.
5. Certified stocks.

Desk notes

Add your own observations here.

Frozen Concentrated Orange Juice (FCOJ)

A tiny, illiquid market with extreme volatility. Brazil (São Paulo) produces most of the world's orange juice; Florida output has collapsed due to citrus greening and hurricanes.

Contract specifications

Contract	Venue / ticker	Size	Tick	Months
FCOJ-A	ICE US OJ	15,000 lb orange solids	0.05¢/lb = $7.50	F H K N U X

Fundamentals

- **Supply:** Brazil (~70%+ of world exports; Fundecitrus crop estimates in May), Florida (greening disease — HLB — cut output by ~90% from peak), Mexico.
- Hurricanes (Florida, Jun–Nov), freezes (Dec–Feb), Brazil drought.
- **Demand:** US, EU; declining per-capita consumption; price-elastic at high levels.
- Large price swings with low liquidity — wide bid/offer and daily limits.

Desk notes

Add your own observations here.

Livestock & Dairy

Cattle (Live & Feeder)

Contract specifications

Contract	Venue / ticker	Size	Tick	Settlement	Months
Live Cattle	CME LE	40,000 lb	0.025¢/lb = $10	Physical delivery	G J M Q V Z
Feeder Cattle	CME GF	50,000 lb	0.025¢/lb = $12.50	Cash vs CME Feeder Cattle Index	F H J K Q U V X

The cattle cycle

- Beef cattle have a long biological lag: cow bred → calf born ~9 months later → weaned at 6–8 months → backgrounded → **feedlot** ~150–180 days → slaughter at ~18–22 months. Heifer retention to rebuild the herd takes ~2 years to add supply.
- **The cycle runs ~8–12 years.** Drought forces liquidation (more beef short term, less later). The US herd fell to its lowest level since the early 1950s in 2024–2025, driving record cattle prices.
- **Heifer retention** paradoxically tightens supply further in the short term, since heifers are withheld from slaughter.

Margins and spreads

- **Cattle crush / feeding margin:** live cattle − feeder cattle − feed (corn) cost. Feeder cattle price is inversely related to corn (feed cost).
- **Boxed beef cutout** (Choice/Select, USDA daily) vs cash fed cattle → packer margin. Packers (Tyson, JBS, Cargill, National Beef) cut slaughter when margins are negative.
- Live cattle futures normally trade at a discount/premium to the 5-area cash fed steer price; basis converges at expiry.

Key reports

- **Cattle on Feed** (monthly): placements, marketings, on-feed numbers.
- Semi-annual **Cattle inventory** (January, July).
- Weekly slaughter, carcass weights, boxed beef, export sales.
- Drought monitor (pasture conditions).

Other drivers

- Imports: Brazil, Australia; Mexican feeder imports (suspended in 2024–2025 due to **New World screwworm**).
- Tariffs on Brazilian beef; retail beef demand and consumer spending.
- Seasonality: grilling season demand (spring) supports cutout; fed cattle supplies tend to rise in summer.

Desk notes

Add your own observations here.

Lean Hogs

Contract specifications

Contract	Venue / ticker	Size	Tick	Settlement	Months
Lean Hogs	CME HE	40,000 lb	0.025¢/lb = $10	Cash vs CME Lean Hog Index (2-day weighted average)	G J K M N Q V Z
Pork cutout	CME PRK	40,000 lb		Cash vs USDA pork cutout	

Fundamentals

- Short biological cycle: gestation ~114 days, market weight ~6 months after birth → supply responds within ~10 months.
- **Strong seasonality:** prices peak in summer (Jun–Jul) as slaughter is lowest and grilling demand is high, and trough in Nov–Dec when slaughter is highest.
- **Hog crush (margin):** hogs − (corn + soybean meal) feed cost.
- Packer margin: pork cutout vs hog prices; plant capacity limits (Saturday kills).

Demand

- US domestic, plus exports to Mexico (#1), China, Japan, South Korea, Canada. China's hog herd (largest in the world; African Swine Fever 2018–2020 halved it) swings import needs. Chinese tariffs on US pork are recurring.
- Variety meats (offal) exports to China add value per head.

Key reports

- Quarterly **Hogs and Pigs** (USDA): inventory, breeding herd, farrowing intentions, pigs per litter.
- Daily slaughter, weights, pork cutout; weekly export sales; cold storage (monthly).
- Disease risk: African Swine Fever (ASF), PRRS, PED.

Desk notes

Add your own observations here.

Dairy

Contract specifications

Contract	Venue	Size	Notes
Class III Milk	CME DC	200,000 lb	Cash vs USDA Class III price (cheese milk)
Class IV Milk	CME DK	200,000 lb	Butter/powder milk
Cheese (blocks/barrels)	CME CSC	20,000 lb	
Butter, NFDM, dry whey	CME	20,000 lb / 44,000 lb	
Whole milk powder (WMP)	SGX-NZX	1 t	Fonterra / GDT benchmark

Fundamentals

- **Global Dairy Trade (GDT)** auction every two weeks (Fonterra-led) sets international powder/butter prices.
- Exporters: New Zealand, EU, US. Importers: China, Southeast Asia, Middle East, Mexico.
- Supply drivers: herd size, feed costs (milk–feed ratio), weather/pasture (NZ season Aug–May), disease (H5N1 avian flu in US dairy herds since 2024).
- US Federal Milk Marketing Orders define class prices (Class I fluid, II soft products, III cheese, IV butter/powder); reform in 2025 changed formulas.

Desk notes

Add your own observations here.

Carbon, Freight & Other Markets

Carbon & Environmental Markets

Major compliance markets

Market	Instrument / venue	Size	Notes
EU ETS	EUA, ICE Endex / EEX	1,000 t CO2	December contract is the benchmark; power, industry, aviation, maritime (phased from 2024)
UK ETS	UKA, ICE	1,000 t	Linking with EU ETS under negotiation
California / Québec	CCA, ICE	1,000 t	Quarterly auctions; floor price
RGGI (US Northeast)	RGA, ICE	1,000 t	Power sector
China national ETS	Shanghai Environment Exchange	—	Power, expanding to steel, cement, aluminium
Voluntary	CBL N-GEO, GEO; Verra/Gold Standard credits	—	Quality and integrity concerns

EU ETS mechanics

- Cap declines annually (linear reduction factor raised under Fit for 55); free allocation phases out as **CBAM** phases in (2026–2034).
- **Market Stability Reserve (MSR)** removes surplus allowances when total number in circulation (TNAC) is high.
- **ETS2** (buildings and road transport) planned from 2027/2028.
- Auction calendar (EEX) and compliance deadline (30 September) shape supply/demand.

Price drivers

1. Fuel switching: gas vs coal costs — higher gas prices raise coal burn and EUA demand.
2. Industrial output (recession lowers emissions).
3. Renewable output (wind, hydro, solar) displaces fossil power.
4. Policy announcements (MSR changes, REPowerEU sales, cap adjustments).
5. Speculative positioning by investment funds (EU publishes weekly CoT).

US RINs and LCFS

- **RINs** (Renewable Identification Numbers) under the US RFS: D6 (corn ethanol), D4 (biomass-based diesel), D5 (advanced), D3 (cellulosic). Obligated parties (refiners) must retire RINs. D4 prices link soybean oil to policy.
- **California LCFS** credits: carbon-intensity-based; price sensitive to rulemaking.

Desk notes

Add your own observations here.

Freight (Dry Bulk & Tankers)

Shipping costs determine arbitrage between regions; freight can be traded via **Forward Freight Agreements (FFAs)** cleared on SGX and EEX.

Dry bulk

Vessel	Size (dwt)	Main cargoes	Benchmark
Capesize	150,000–200,000+	Iron ore, coal	Baltic Capesize 5TC; C3 (Tubarão–Qingdao), C5 (W. Australia–Qingdao)
Panamax / Kamsarmax	65,000–85,000	Coal, grains	Baltic Panamax 5TC/4TC
Supramax / Ultramax	50,000–65,000	Grains, fertiliser, minor bulks	Baltic Supramax 10TC
Handysize	25,000–40,000	Minor bulks	Baltic Handysize 7TC

The **Baltic Dry Index (BDI)** is a weighted composite of these.

Tankers

Vessel	Size	Use
VLCC	~2 million bbl	Long-haul crude: Middle East Gulf–China (TD3C), US Gulf–China
Suezmax	~1 million bbl	West Africa, Black Sea, US Gulf to Europe
Aframax	~600–750k bbl	Short-haul crude: North Sea, Med, Caribbean
LR2 / LR1 / MR		Clean products

Rates quoted in **Worldscale** points (percentage of a notional flat rate per route) or $/day time-charter equivalent (TCE).

Drivers

1. Tonne-miles: the distance cargoes travel (e.g., Red Sea diversions around the Cape from late 2023 raised tonne-miles sharply).
2. Fleet supply: orderbook, scrapping, IMO environmental rules (CII, EEXI), sanctioned "shadow fleet".
3. Seasonal cargo flows: Brazil soy (Q1–Q2), US Gulf grain (Q4), Australian ore (cyclones Q1).
4. Port congestion, canal restrictions (Panama Canal drought 2023–2024).
5. Bunker fuel prices.

Desk notes

> Add your own observations here.

Lumber, Fertilisers & Other Niche Markets

Lumber

- **CME Lumber (LBR):** 27,500 board feet of 2×4 softwood, physically delivered (launched 2022 replacing the illiquid random-length contract). Thin liquidity.
- Drivers: US housing starts and repair/remodel demand, Canadian supply (BC mill closures, wildfires, pine beetle), **US duties on Canadian softwood lumber**, interest rates.

Fertilisers

- **Nitrogen** (urea, ammonia, UAN): made from natural gas → European gas prices set the marginal cost. Chinese export restrictions and Indian import tenders are key.
- **Phosphates** (DAP, MAP): Morocco (OCP), China (export quotas), US (Mosaic), Saudi Arabia (Ma'aden).
- **Potash** (MOP): Canada (Nutrien), Russia, Belarus — highly concentrated; annual China/India contract prices set the tone.
- Venues: CME urea/DAP/UAN swaps (cash-settled vs Argus/CRU/Fertilizer Week). Fertiliser affordability influences corn acreage and yield.

Other markets to consider adding

- **Rubber** (SGX TSR20, OSE RSS3, SHFE) — Thailand/Indonesia supply; auto tyre demand.
- **Wool, potatoes (EEX), cheese, salmon (Fish Pool / Euronext)**.
- **Rare earths** (NdPr oxide) — Chinese supply and export controls.
- **Molybdenum, vanadium, tungsten, antimony, gallium, germanium** — critical minerals subject to Chinese export controls.
- **Hydrogen, RECs/GOs, weather derivatives (CME HDD/CDD)**.

Desk notes

> Add your own observations here.

Appendix

Conversions & Glossary

Unit conversions

From	To	Factor
1 metric tonne	lb	2,204.62
1 short ton	lb	2,000
1 troy oz	grams	31.1035
1 barrel	US gallons / litres	42 / 158.987
1 tonne crude (avg)	barrels	~7.33 (varies with density)
1 tonne gasoil	barrels	~7.45
1 tonne gasoline	barrels	~8.45
1 MWh	MMBtu	3.412
1 therm	MMBtu	0.1
1 tonne LNG	MMBtu	~52
Bushels per tonne	corn / wheat & soy / oats	39.37 / 36.74 / 68.89
¢/lb → $/t	multiply by	22.0462
1 bale cotton	lb	480
1 bag coffee	kg	60

Glossary

Term	Meaning
Backwardation	Prompt price above deferred price; signals scarcity
Basis	Cash price minus futures price at a location
Contango	Deferred price above prompt; signals surplus/storage economics
Convenience yield	Implicit benefit of holding physical inventory
Crack spread	Refining margin (products minus crude)
Crush spread	Processing margin (soy products minus beans; hogs/cattle minus feed)
Differential	Premium/discount of a physical grade/origin vs benchmark
EFP	Exchange for physical — swaps a futures position for a physical one
FND	First notice day: earliest date delivery notices may be issued
LTD	Last trading day
Open interest	Number of open futures contracts
OSP	Official selling price set by national oil companies
Spark/dark spread	Power margin over gas/coal generation
Stocks-to-use	Ending stocks ÷ annual use; key ag tightness metric
TC/RC	Treatment/refining charges paid by miners to smelters
Warrant	Title document for metal in an exchange warehouse
WASDE	USDA World Agricultural Supply and Demand Estimates

Desk notes

Add your own observations here.

Trading Journal & Running Notes

Use this chapter for dated market observations, trade ideas, post-mortems, and anything that does not fit a single commodity chapter. Newest entries at the top.

Template

> **Date:** YYYY-MM-DD · **Market:** · **View:** · **Catalyst:** · **Invalidation:** · **Outcome / lesson:**

Entries

- *(No entries yet.)*